

# Om Mandhane

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## EDUCATION

**Vivekanand Education Society's Institute of Technology, Mumbai**

2022 – 2026

*Bachelor of Engineering in Automation & Robotics*

*Minor in Artificial Intelligence and Machine Learning | CGPA: 8.65/10*

## PUBLICATIONS

**Robust 2D Traversability Mapping for Construction AMRs via Failure-Mode-Aware Fusion of LiDAR Geometry and Monocular Semantics.** Manoj Karnekar, **Om Mandhane**, Gautham Ramkumar. *Submitted to IEEE IROS 2026 Workshop FoC via OpenReview.*

**Phone2Act: A Low-Cost, Hardware-Agnostic Teleoperation System for Scalable VLA Data Collection.** **Om Mandhane**, Bipin Yadav, Sangeetha Prasanna Ram, Gopalakrishnan Narayanan. *Submitted to IEEE ICMRE.* [arXiv] [Website] [Github]

## EXPERIENCE

**Flo Mobility, Bengaluru, India**

Jun 2026 – Aug 2026

*Robotics Engineering Intern*

Evaluated and deployed state-of-the-art monocular metric depth models (Metric3D v2, UniDepth) on an NVIDIA Jetson AGX Orin utilizing ONNX and FP16 TensorRT, enabling real-time point cloud generation for a custom downstream navigation pipeline.

Engineered a real-time multimodal perception stack in ROS 2, fusing 3D Livox LiDAR geometry with a custom 28-class YOLOv26 segmentation model. Developed a failure-mode-aware deterministic fusion engine to successfully mitigate geometric false positives (ramps) and false negatives (wet mud).

Enabled the fusion architecture by building a SAM 3-assisted annotation workflow to curate a multimodal construction dataset.

**Bhabha Atomic Research Centre (BARC)**

Dec 2024 – Mar 2025

*Project Intern*

*Developed a ROS2-based software stack for a custom indoor mobile robot, including URDF creation, Gazebo simulation, and real-time hardware interfacing. Also integrated and tuned the Nav2 stack for autonomous navigation in simulation.*

*Implemented and optimized the VFH algorithm on both simulation and the real robot for dynamic object avoidance and navigation.*

**ROS 2 & Gazebo Workshop Instructor (ISA VESIT)**

Dec 2024 & Jun 2025

*Technical Instructor*

*Led two 3-day workshops on ROS 2 and Gazebo for undergraduate students, conducted during winter and summer breaks, in collaboration with ISA VESIT.*

**Vivekanand Education Society's Institute Of Technology**

May 2024 – Jul 2024

*Summer Intern*

*Engineered a ROS 2 UR5 manipulation pipeline in Gazebo using the TRAC-IK solver, and integrated a custom C++ pose-input interface for command-line end-effector control.*

## PROJECTS

**Phone2Act: A Low-Cost, Hardware-Agnostic Teleoperation System for VLA Data Collection**

Oct 2025 – Mar 2026

Developed a **hardware-agnostic 6-DoF ROS 2 teleoperation framework** transforming smartphones into spatial controllers via ARCore, addressing the data bottleneck in Vision-Language-Action (VLA) robotics through scalable demonstration collection.

Designed a modular three-layer architecture (interface, planner, hardware bridge) decoupled via standardized ROS 2 topics; implemented workspace clamping and a **zero-jump filter** for safe real-robot deployment.

Built a **Universal Recorder** synchronizing multi-camera streams with robot state at 20Hz and exporting directly to **LeRobot dataset format** (MP4/Parquet), creating a turnkey pipeline from teleoperation to VLA training.

Validated framework across single-arm (Dobot CR5) and bimanual (LeRobot SO-101) platforms with **395ms end-to-end latency**; collected 130 episodes to fine-tune NVIDIA's **GR00T-N1.5-3B** diffusion policy using **10-step temporal action chunking** and deployed via cloud-based inference bridge.

**Autonomous Navigation with Model Predictive Control (MPC) [Github] [Video]**

Apr 2026 – May 2026

Engineered a real-time ROS 2 local planner using **CasADi/IPOPT** to solve a Receding Horizon optimization problem, enforcing **Uni-cycle kinematic constraints** to guarantee physically executable trajectories.

Formulated dynamic obstacle avoidance via an **Inverse Barrier cost function**, downsampling 2D LiDAR scans to map physical surface contours and generate repulsive potential fields.

Pre-processed sparse waypoints into  $C^2$ -continuous reference paths using **SciPy cubic B-splines** and implemented a **Monotonic Progress Tracker** window to prevent trajectory path-jumping during sharp hairpin maneuvers.

Designed a **Cruise Control cost function** to heavily penalize deceleration and enforce forward progression, while eliminating saddle-point decision oscillations by injecting **solver warm-starting** inertia.

**HexBot [GitHub] [Video]**

Sep 2024 - Mar 2025

Developed an autonomous differential drive robot using **SLAM Toolbox** for real-time mapping and **Nav2** for path planning, alongside a VR-based video streaming interface for remote monitoring.

**Warehouse Drone [Video]**

Sep 2024 - Feb 2025

As part of the e-Yantra Robotics Competition by IIT Bombay, we built an autonomous drone system in a simulated warehouse using **ROS2** and **Gazebo**; implemented **RRT\*** for dynamic path planning and computer vision for package recognition. Tested various **PID tuning methods** including Threshold-Based Gain Scheduling and integral windup protection to enhance stability and ensure accurate stopping in front of packages.

**GeoGuide [GitHub] [Video]**

Aug 2023 - Feb 2024

Developed an autonomous three-wheeled robot for the IIT Bombay **e-Yantra Robotics Competition**, guided by an overhead camera system. Applied **AlexNet-based transfer learning** for disaster image classification and used **Dijkstra's algorithm** for priority-based navigation between event locations. Integrated **ESP32 Wi-Fi communication** and QGIS-based localization.

SKILLS

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- Programming:** C++, Python, Embedded C
- AI & Perception:** PyTorch, TensorFlow, OpenCV, YOLO, CNNs, Transformers, Diffusion Models, Monocular Depth Estimation, TensorRT, ONNX, Vision-Language-Action (VLA) Models
- Robotics:** ROS 2, MoveIt, Nav2, SLAM Toolbox, Gazebo, Path Planning, PID, MPC
- Libraries:** NumPy, Pandas, scikit-learn
- Tools:** Git, GitHub, Linux CLI, VS Code, Jupyter, Google Colab, MATLAB, Simulink
- Hardware:** NVIDIA Jetson AGX Orin, Arduino, ESP32, Raspberry Pi | **Training Certificates**

AWARDS & HONORS

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- Stage 2 Qualifier, **IIT Bombay e-Yantra Robotics Competition** — 2023-24 and 2024-25
- Awarded **ISA General Education Endowment Scholarship (\$2,000)** by ISA for outstanding potential in automation — 2025
- Recipient of **Tecnimont (MAIRE) Scholarship** for academic excellence — 2024-25
- Qualified **Prototype Stage (Stage 3), Vishwakarma Awards 2025** — Smart Mobility — 2025
- 7th Place**, India Automation Games — 2024
- Cleared **GATE 2026**(CS, Data Science & AI)